MILITARY SPECIFICATION

TUBE, BIOLOGICAL CULTURE SAMPLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of sterile, disposable, plastic culture tube with two applicators, suitable for obtaining and maintaining uncontaminated cultures.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

JJJ-C-561	Cotton, Sterile (Purified) and Nonsterile.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-S-36586	Sterilization Test Strip Set, Bacterial Spore.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-1363	Measurement of Wood Moisture Content.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as hereinafter specified.

3.1.1 Wood. Wood shall be tough, flexible and well-seasoned and shall be free of cross grain, knots, stains and fungus growth. Moisture content shall be not less than 6 percent and not more than 12 percent when tested as specified in 4.4.1.

3.1.2 Cotton. Cotton shall conform to JJJ-C-561, Grade A, except the sterility requirements shall not apply.

3.1.3 Polyethylene terephthalate. Polyethylene terephthalate shall be free of optical bleach.

3.2 Construction. The culture tube shall be provided with a tube, a cap and two applicators enclosed in a bag.

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3.2.1 Bag. Bag shall be fabricated of paper or plastic, shall completely enclose the tube, and shall be sealed by glue or by heat. The bag shall be easily opened and shall maintain the sterility of the tube and contents until opened. In addition, each bag shall be marked with lot control number and the word "Sterile" and the legend: "Sterility guaranteed unless package has been opened or damaged."

3.2.2 Tube. Tube shall be fabricated of clear polystyrene with one end open and one end closed. The diameter of the open end shall be greater than the closed end so as to form a tube with a uniform taper. The outside surface near the open end shall have a flange or screw threads. The bottom of tube may be flat or rounded.

3.2.3 Cap. Cap shall be polyethylene and shall be designed and attached so as to permit sterilization of contents of the tube. The inside of the cap shall have two sockets. The inside surface of the side wall shall contain a notch or screw threads with the flange or screw threads on the tube to form a secure closure. Culture tube may be furnished with the cap and tube not completely closed, provided the applicators are inside the tube and the tube can be removed from the bag by grasping the cap.

3.2.4 Applicators. Applicators shall be smooth wood or plastic sticks with pear-shaped cotton or polyethylene terephthalate tips secured to and extending at least 1/16-inch beyond one end of the stick. Opposite ends of the sticks shall be secured to the sockets in the cap. Cotton tips may be secured by glue and 1000 cotton tips shall weigh 600 $\pm$ 200 grains. If plastic, the applicator stick shall have a bending arc of not less than 80 degrees, and shall show no evidence of fracturing, splitting or breaking when tested as specified in 4.4.4. Plastic applicator sticks, if used, may be scored to facilitate breaking of the stick. Instructions (see 3.6) shall identify the scoring technique and breaking procedure.

3.3 Sterility. The biological culture sampling tube shall be sterile when tested with a biological indicator as specified in 4.4.2.

3.3.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. National Stock Number.
c. Contract or purchase order number.
d. Sterility lot or control number.
e. Date of sterilization.

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f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.)
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.3.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.4 Toxicity. Applicator and tube shall not be toxic when tested as specified in 4.4.3. Growth for all samples shall be heavy or good and shall be as great or greater than the control.

3.5 Dimensions shall be as shown below:

Table I Dimensions (in inches)

Length of assembled tube and cap	6-3/4 Max.
Length of tube	5-3/4 ± 5/16
Distance from bottom of cap to end of applicator	4-1/2 Min.
Diameter of applicator stick	0.084 ± .004
Distance end of tip of applicator to bottom of tube	1/4 + 1/8, - 3/16
Length of applicator tip	5/8 ± 1/16
Outside diameter of applicator tip	3/16 ± 1/32
Outside diameter 1/4 in. (or less) from bottom of tube	5/8 + 1/16, -1/8
Outside diameter immediately below flange (flange type tube)	11/16 ± 1/16
Outside diameter at top of tube (screw thread tube)	5/8 ± 1/16
Wall thickness of tube	0.020 Min.

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3.5.1 Tolerances. Tolerances indicated above are the maximum permissible for a single dimension. Where the use of a tolerance would result in an item which does not conform with the requirements of other portions of this specification, tolerance shall not be permitted and other portions of the specification shall govern.

3.6 Instructions. If scored plastic applicator sticks are supplied, instructions shall be provided identifying the scoring technique and breaking procedure (see 3.2.4). These instructions may be supplied with instructions specified in 5.1.2.1.

3.7 Workmanship. Workmanship shall be first class throughout. Units shall be free from defects listed in Table IV and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the cotton, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual tests results and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. For inspection purposes the unit of product shall be the culture tube.

Table II Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major A defects	II	2.5
Major B defects	S-1	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. Acceptance number shall be zero.

Table III Sampling for tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Wood applicator	Moisture content of wood	3.1.1	4.4.1	*
Tube	Sterility test	3.3	4.4.2	USP
Tube	Toxicity test	3.4	4.4.3	S-1
Plastic applicator	Bending arc	3.2.4	4.4.4	S-1

* Two sticks of wood or plastic from each lot used in the manufacture of the applicators shall be selected for testing (4.4.1).

4.3 Examination. The biological culture sampling tubes shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in Table IV.

Table IV Classification of defects

Category	Defect
Major A	
101	Component not free of deformation.
102	Bag not free from tears.
103	Bag not sealed.
104	Cap not properly secured to tube.

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Classification of defects (cont'd)

Major A	Defect
105	Cap and tube not free of cracks.
106	Tube not clear.
107	Applicators not secured in sockets.
108	Applicators not free of stains and knots.
109	Applicators not free of cracks and slivers.
110	Cotton tips not securely wound around the applicators.
Major B	
151	Instructions not present.
152	Instructions not complete, correct and legible.
Minor	
201	Marking not complete, correct and legible.

4.3.1 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Moisture content of wood. The moisture content of the wood shall be determined as specified in MIL-STD-1363.

4.4.2 Sterility test. The biological culture sampling tubes shall be tested for sterility in accordance with the method specified in the USP for non-parenteral devices using inoculated product or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.2.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table III.

4.4.2.1 Inoculated product. For each type of sterilization, the organism used shall be as listed below:

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Type of Sterilization	Organism	Incubation temperature
Steam	B. stearothermophilus	55°-65°C.
Dry heat	B. stearothermophilus, or	55°-65°C.
	B. subtilis var. niger (or globigii)	30°- 35°C.
Gas	B. subtilis var. niger (or globigii)	30°- 35°C.
Radiation	B. pumilus	30°- 35°C.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.2.2 Inoculated carrier. Inoculated carriers (paper spore strips), if used, shall be inoculated to be of equivalent resistance to spore populations included in MIL-S-36586. The method of incubation and number of test units shall be as indicated in 4.4.2 and 4.4.2.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.2.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.2.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a Government contract.

4.4.3 Toxicity.

Cultures

Prepare 24-hour broth cultures of:

a. Beta hemolytic streptococcus
b. Staphylococcus aureus

By means of sterile applicator, inoculate each culture on a trypticase soy blood agar plate (see 6.1) and incubate agar plate at 32°Centigrade (°C.) for 48-72 hours.

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Control

Control tube shall be similar in dimensions to the specified bacteriological tube (see 3.5). Dimensions shall be such that the applicator tips of the control are submerged to the same degree (± 5 percent) as that of the sample. Bag (see 3.2.1) need not be a part of the control. For the control, glass shall be substituted for the plastic tube, and metal for the plastic cap and wood applicator. USP cotton shall be used for the applicator tip. Tip shall be secured to the applicator without the use of adhesive or other foreign material. All surfaces on the applicator and inside surfaces of the cap and tube shall be clear and free of foreign material. Control tube shall be sterilized by autoclaving at 15 pound pressure and 121°C. for 1/2 hour with the cap and the applicator in place.

Procedure

a. One control shall be used with each test sample.
b. Inoculate different applicator from each plate culture. Rotate the side of the applicator tip one full turn to cover approximately one square centimeter of the plate culture surface.
c. Replace the applicator in its own tube.
d. Incubate the inoculated applicator tips for 24 hours at 32°C.
e. Add 2 ml. of sterile Trypticase Soy Broth (see 6.1) into each tube. Broth shall immerse the applicator tip.
f. Incubate the tubes and applicator tips for two hours at 32°C. Tubes shall be in an upright position.
g. Remove applicator and inoculate to an agar plate (see 6.1).
h. Incubate agar plates at 32°C. for 48-72 hours.
i. Examine and evaluate growth as trace, poor, good or heavy.

4.4.4 Bending arc. Remove the plastic applicator stick from the cap, hold it firmly between the fingers, gripping it approximately one inch from each end. Gently bend the applicator stick through an arc. Observe the angle at which the applicator fractured. If stick is scored and scoring is more than 1 inch from the end, stick may be gripped at the score.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

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Table V Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One package containing one hundred culture tubes, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. One hundred tubes, individually packaged as specified in 3.2.1, along with two copies of directions for use, shall be further packaged in a box of appropriate size and constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure of the box shall conform to method V as specified in PPP-B-636.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Six unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to

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applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.2 Level B. Six unit packages shall be packed in a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in the appendix of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Sterility lot (control) number, contract or purchase order number and name of contractor shall also be shown. The method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and the date of packaging. EXAMPLE: "A-1/75-G." NOTE: Final sterilization shall be on or after indicated packaging date.

5.3.2 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. Sterility lot (control) number shall be shown. The method of sterilization as specified in 5.3.1 shall also be shown.

6. NOTES

6.1 Trypticase Soy blood agar plates and Trypticase Soy Broth may be obtained from Baltimore Laboratories as BBL 168 and BBL 162 respectively.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-729-6484	TUBE, BIOLOGICAL CULTURE SAMPLING, Plastic, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6640-0924
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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-T-37193, Tube, Biological Culture Sampling

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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